



EXTREME HEAT PLAN

INDIGO SHIRE
2025 - 2028



This complementary plan was adopted by Indigo Shire Council CEO on 28 July 2025 and endorsed by the Indigo Municipal Emergency Management Planning Committee at their meeting on 6 August 2025. The Final Plan was approved by the CEO Trevor Ierino on 25 August 2025.

Version Control table

Version Number	Date of Issue	Author(s)	Brief Description of Change
Version 1	25/7/10	Mark Greene	Draft doc development
Version 2	01/07/16	Karen van Huizen	Update data and minor amendments
Version 3	01/03/2018	Karen van Huizen	Complete review, including addition of heat threshold and change in Lead agency and update of communication plan
Version 4	04/06/2019	Chris Rule & Eloise Martin (DHHS)	Complete Review
Revised Version 5	28/07/2025	Craig McClanahan Anton Mass Nadia Corsini Christy Jackson – Dept of Health	Revision Completed

Amendment Record

Version	Date	Page(s) amended	Description of Change	Person Responsible
3	25/3/2019	5	Inclusion of State Heatwave Plan descriptions	Chris Rule
3	25/3/2019	18	Amendment to Power & Water effects	Chris Rule
3	25/3/2019	19	Inclusion of North East Water as partner agency & inclusion of actions by energy & water providers	Chris Rule
4	4/6/2019	Various	Complete Review	Chris Rule & Eloise Martin
4	19/8/2019	22	Removal of Awareness Campaign in Stage 1	ISC Exec. Management Team
4	12/9/2019	11	Insertion of paragraph relating to climate change	Chris Rule
5	11/07/2025	Various	Complete revision of plan and amended to current legislation and terminology	Craig McClanahan

Contents

1	WHAT IT IS ALL ABOUT	4
2	WHY WE NEED AN EXTREME HEAT PLAN	5
3	EXTREME HEAT PLANNING	6
4	WHAT OUR COMMUNITY LOOKS LIKE.....	7
	Our Setting	7
	Our Population	7
	Our lifestyle	7
	Our Climate.....	8
	Climate Change.....	8
5	HOW HEAT HEALTH AFFECTS THE INDIGO SHIRE.....	9
6	DEFINITION OF HEATWAVE.....	10
7	HOW HEAT WAVES AFFECT OUR HEALTH.....	12
	Prevention	13
8	RECOMMENDED COOL RELIEF SPACES	13
9	PEOPLE MOST AT-RISK TO HEATWAVE CONDITIONS	14
	Our elderly.....	14
	Our young children.....	15
	People with chronic disease	15
	People with a disability.....	15
	People on medication.....	16
	People who are outdoors	16
	Our tourists.....	16
	Our farmers	16
	Likely impacts	17
10	OTHER EFFECTS OF EXTREME HEAT.....	17
	Power	17
	Water.....	17
	Animals	18
11	OUR PARTNERS.....	19
12	WHAT WE ARE GOING TO DO	19
	Our action plan	19
13	HOW WE EVALUATE THE PLAN’S EFFECTIVENESS.....	21
14	Appendices	22
	Appendix 1: Key terms.....	22
	Appendix 2: Heatwave action & communications plan	23
	Appendix 3 – Flows Chart for Communications Plan	28

1 WHAT IT IS ALL ABOUT

As we face a future with an increased likelihood of hot weather, it is important that the Municipal Emergency Management Planning Committee Indigo Shire, continues to develop, maintain and implement a municipal-wide approach to dealing with heatwave events. This plan provides a framework of support to our local community and vulnerable population groups so we can be better equipped to handle more intense and frequent heatwaves.

The State Emergency Management Plan, Extreme Heat Sub-Plan 2022, sets out arrangements to ensure an integrated and coordinated approach to the management of heat events, to reduce the impact and consequences of extreme heat on the community, infrastructure and services. It outlines responsibilities, a collaborative approach to be taken by government and emergency management agencies, and planning and arrangements to be made by functional sectors of government and industry to prepare for a heat event to reduce its impact on the community, infrastructure and services. This plan can be found at

<https://www.emv.vic.gov.au/responsibilities/state-emergency-management-plan-sub-plans/sempp-extreme-heat-sub-plan>

The State Extreme Heat Sub-Plan distinguishes between single days of extreme heat and prolonged heatwaves. Extreme heat occurs when the forecast average temperature on any day exceeds the predetermined heat health temperature threshold in a Victorian weather forecast district. A heatwave involves three or more consecutive days of extreme heat.

The Extreme Heat Plan for Indigo Shire is also underpinned by a long-term approach by the Planning Committee, in partnership with stakeholders and neighbouring shires, to reduce the scale of climate change and decrease its overall impact on the local environment.

The *Plan* has been written to:

- Be consistent with departmental guidelines and other Council planning frameworks;
- Outline health, community and emergency service actions and response arrangements to heatwave alert;
- Identify vulnerable persons within the Indigo community;
- Propose a clear communication strategy to initiate alert, response and recovery phases of the plan;
- Describe key stakeholder roles and responsibilities; and
- Promote a community awareness and education component.



2 WHY WE NEED AN EXTREME HEAT PLAN

The Municipal Emergency Management Planning Committee have undertaken a community risk assessment and identified heat to be a key risk for Indigo Shire. In the State Extreme Heat Sub-plan, it encourages MEMPCs to prepare at an level extreme heat plan. The plan is designed to support their local communities to adapt to heatwave conditions. By developing our own plan, we can use our understanding of local conditions and resources to better prepare for, respond to and recover from heatwave conditions.

High temperatures can seriously impact on the health of people in our shire, especially the very young, the elderly, people with chronic disease, those with a disability, on medication or working outdoors, tourists and farmers. We need to be well prepared in advance to make sure our community has the necessary tools to cope in times of extreme heat.

Projected climate change impacts for the region indicate increasing frequency and duration of heatwaves in the future.

By having an *Extreme Heat Plan*, we can:

- Ensure that health information and support is readily available to our community;
- Increase the capacity of our community to respond during extreme heat days;
- Manage an extreme heat emergency more effectively; and
- Develop long term changes in our behaviour to improve our health and wellbeing.

The aim of our *Extreme Heat Plan* is to:

- Support the Indigo community to prepare-for, respond-to and recover-from extreme heat conditions.

We will achieve this by:

- Identifying vulnerable population groups in our community and the risks they face during extreme heat;
- Developing partnerships with local organisations to better coordinate a response to extreme heat
- Outlining effective strategies and actions to implement in the event of a extreme heat; and
- Building practices to evaluate the ongoing effectiveness of the plan.

3 EXTREME HEAT PLANNING

All Victorian councils are required to strive to meet the goals outlined in the *Extreme Heat Sub-Plan 2022* which is a sub plan of the *State Emergency Management Plan (SEMP)*. These goals include:

- Councils are encouraged to monitor weather conditions and respond in accordance with their extreme heat plans on receipt of a Bureau of Meteorology Heatwave Warning or a Chief Health Officer Heat Health warning ;
- Contact agencies responsible for at-risk Victorians listed on the Vulnerable Person's Register (VPR), to ensure their safety and wellbeing during a heatwave; and,
- Use local media and community services networks to inform and convey important information and advice on staying healthy in the heat to the Indigo shire community.

The *Indigo Shire Extreme Heat Plan* is a complementary plan of the *Municipal Emergency Management Plan (MEMP)* and is used by Council and its partners in planning-for and responding- to extreme heat conditions. It has clear links to the *Indigo Shire Municipal Public Health and Wellbeing Plan* and the *Council Plan*.

The plan has been developed using the general principles of emergency management as a guide to planning, preparation, response and recovery under the Victorian Government legislative framework relating to heat events detailing the different areas across government and the health and community sectors. This includes:

- *Emergency Management Act 1986;*
- *Planning and Environment Act 1987;*
- *Local Government Act 2020;*
- *Public Health and Wellbeing Act 2008;*
- *Emergency Management Act 2013; and,*
- *Climate Change Act 2017.*
- *State Emergency Management Plan – Extreme Heat Sub-Plan 2022*

A Community Emergency Risk Assessment (CERA) has been completed for heatwaves and is found in 4.5.5 of the *Indigo Shire Council Municipal Emergency Management Plan*.



4 WHAT OUR COMMUNITY LOOKS LIKE

Our Setting

The Shire of Indigo is named after the Indigo Valley and Indigo Creek. It is in the North East of Victoria, about 300 kilometres north-east of Melbourne. Our setting is bounded by the Murray River and adjoins the following municipalities:

- Wodonga City Council to the north
- Towong Shire Council to the northeast
- Alpine Shire Council to the east
- Rural City of Wangaratta to the south

The shire is predominantly rural in nature with supporting services in small towns. The mix of historic towns, grape production, high altitude fruit growing and productive river flats has produced a diverse combination of land uses and attributes. Our economy relies on its rural and tourism base and its regional association with Albury-Wodonga and Wangaratta.

Our Population

Indigo Shire had a population of 17,662 as of 2025, we have experienced steady growth since 1981. The future regional population growth is expected to be in the Indigo Shire towns, with the majority expected in Rutherglen and Kiewa-Tangambalanga.

In 2021, the median age of people living in Indigo was 47 years.



Our lifestyle

The natural environment has a significant influence on our lifestyle. The area offers a unique mix of attractive rural surroundings, welcoming historical villages and four distinct seasons that attracts visitors to the area. Many residents and visitors enjoy an outdoors lifestyle, in nature, swimming, cycling, at food and wine or arts and cultural events.

Indigo has significant cycling opportunities with the Murray to mountains rail trail, Beechworth Mountain Bike Park. The inland waterways such as Lake Moodemere, Lake Sambell and Woolshed Falls are linked to many walking tracks, great for a picnic or swimming in the warmer months. Mt Pilot National Park provides opportunities for bushwalking, hiking, bird watching and photography. Winter and summer outdoor sports are also popular within the Indigo shire.

Indigo has many strong vibrant communities, developed around village settings, which is a backdrop for many community events and celebrations. There are more than 1,500 heritage buildings within the shire and a broad range of events that attract thousands of locals and visitors each year to enjoy the region's food, wine, culture and heritage. With people becoming more conscious about food miles and how their food is produced, agritourism is growing in Indigo seeing farmers and producers out at many events showcasing and selling their produce.

Indigo residents have access to a variety of services, healthcare and education around the shire. Many residents commute to Wangaratta or Albury/Wodonga for work, shopping, sporting and entertainment such as cinemas, shows and music events.

Our Climate

Currently, climate in the Indigo area is typified by daily mean summer temperatures ranging from a minimum of 13°C to a maximum of 27°C. Mean daily winter temperature ranging from a minimum of 2°C and a maximum of 9°C. Frosts can occur in Indigo between April and October. Most rainfall occurs in winter and spring. Summer thunderstorms are common, contributing to the risk of fires and floods. On average there are 81.5 days each year where at least one millimetre of rain falls. Climate projections indicate a hotter, drier future with increasing daily mean temperatures.



Climate Change

Climate change is a real and urgent threat to our environment, economy, health and lifestyle. Climate change is likely to exacerbate many of the known disaster risks and affect those already especially vulnerable to natural hazards.

Expectations for hotter and generally drier conditions in Indigo Shire, with increasing frequency and intensity of extreme climatic events, have implications for Emergency Management Planning. Adaptation is necessary in considering a range of council services. In Emergency Management, it is important that all risk evaluation and planning is considered through a future climate lens.

5 HOW HEAT HEALTH AFFECTS THE INDIGO SHIRE

In January 2014, Victoria experienced a significant heatwave. In terms of intensity and duration it was like the extreme heat recorded in January 2009, where the Indigo Shire, along with the rest of Victoria, experienced temperatures among the highest ever recorded. The Chief Health Officer estimated that 374 additional deaths occurred state-wide during this heatwave. Indigo Shire experienced heat of unprecedented intensity and duration with maximum temperatures 12 to 15°C above normal for much of the area. The temperature was above 43°C for three consecutive days from 28 to 30 January reaching a peak of 45.1°C on 30 January 2009.

Research tells us that climate change is expected to increase the frequency and intensity of such heatwaves in our area. During the last decade, average temperatures in our region have been recorded as 0.5°C warmer than the 30-year (1961 to 1990) average. The future climate of our area is expected to be hotter and drier than it is today. By 2030, average annual temperatures will be around 0.9°C warmer and the number of hot days (days over 30°C) will continue to increase.

Statistics also show a decline in the region's rainfall over the past decade. Between 1998 and 2007, our average rainfall was 12% below the 1961 to 1990 average. There were 18 fewer rainy days recorded each year on average. It is predicted that the number of hot days will continue to increase, and rainfall totals will continue to drop. As these predictions in relation to climate change are that extreme weather events will occur more frequently and severely in the years to come, we have a duty to ensure that all residents and services of Indigo are well prepared to protect the most vulnerable members of our communities during such events.

Indigo Shire's diverse landscapes and climate make it vulnerable to drought, bushfire and heatwave conditions. High temperatures in the Indigo area typically correspond to low humidity values because the prevailing winds originate from the dry inland parts of the state. While more humid conditions can exist when the air temperature is closer to 30°C and can provide some degree of discomfort, such days are not generally associated with heatwave conditions. As a result, for Indigo the air temperature alone can provide a reasonable measure of heat stress.

Very hot days are usually associated with hazardous fire weather conditions in the region. The fire threat is increased when thunderstorms develop causing lightning that provides the ignition for fires. This is particularly the case when thunderstorms cause little or no rain, as is sometimes the case in our shire. Following the bushfires of February 2009 and December 2015, Indigo Shire Council received funding from several State Government agencies to rebuild those communities affected by the fires.

The *Indigo Extreme Heat Plan* is underpinned by the Indigo Shire Council Plan where in 2019 council has committed to a NET ZERO BY 2035. These actions were informed by our Emissions Reduction Plan (ERP) in 2021. Indigo Shire Council also works in partnership with the Goulburn Murray Climate Alliance, to reduce the scale of climate change and decrease its impact by adapting the built environment.

In October 2021, Council also signed up to the Cities Power Partnership and Council supports climate actions in our community and we are fortunate to have several trailblazing community sustainability groups such as; Totally Renewable Yackandandah, Totally Renewable Beechworth and Totally Renewable Indigo North and including several local Landcare groups helping to support Council's goal of Net Zero by 2035.

The Indigo Environment Advisory Committee has been invaluable in providing advice to council on environmental matters since 2018.

<https://climateaction.indigoshire.vic.gov.au/>

6 DEFINITION OF HEATWAVE

In Victoria extreme heat plans are activated by local councils when temperatures forecast is likely to impact on the health of the community. These activation levels are also called '*heat health temperature thresholds*'. A heatwave is a complex phenomenon resulting from a certain combination of temperature, humidity, air movement and duration. Heatwaves can mean different things to different people, with thresholds changing according to demographic and geographic circumstances.

Various heat or thermal comfort indices have been developed to evaluate heat-related stress combining air temperature and humidity, and in some cases, wind and direct sunlight.

A heatwave occurs when unusually high maximum and minimum temperatures persist for at least 3 days at a specific location. Temperature thresholds for heatwaves vary and are relative to a location's recent weather and local climate. Alongside unusually high daily maximum temperatures, high overnight temperatures are an important component of a heatwave.

Source: <https://www.acs.gov.au/pages/hazards-heatwaves>

There are several types of heatwaves, based on intensity:

Low-intensity heatwaves are frequent during summer. Most people can cope during these heatwaves.

Severe heatwaves are less frequent. They are likely to be more challenging for vulnerable people. This can include older people, particularly those with medical conditions.

Extreme heatwaves are rare. They are a problem for people who don't take precautions to keep cool – even for healthy people. Anyone who works or exercises outdoors can be at risk.

Source: <http://www.bom.gov.au/australia/heatwave/knowledge-centre/understanding.shtml>

In Indigo the heatwave threshold is set at a mean temperature of 32°C (see figure XX below), over a 24-hour period, which is determined by calculating the daily maximum temperature and the overnight minimum and dividing them by two.

An example of how this calculation made is:

Tuesday

Min: 23°C

Max: 45°C

Wednesday

Min: 26°C

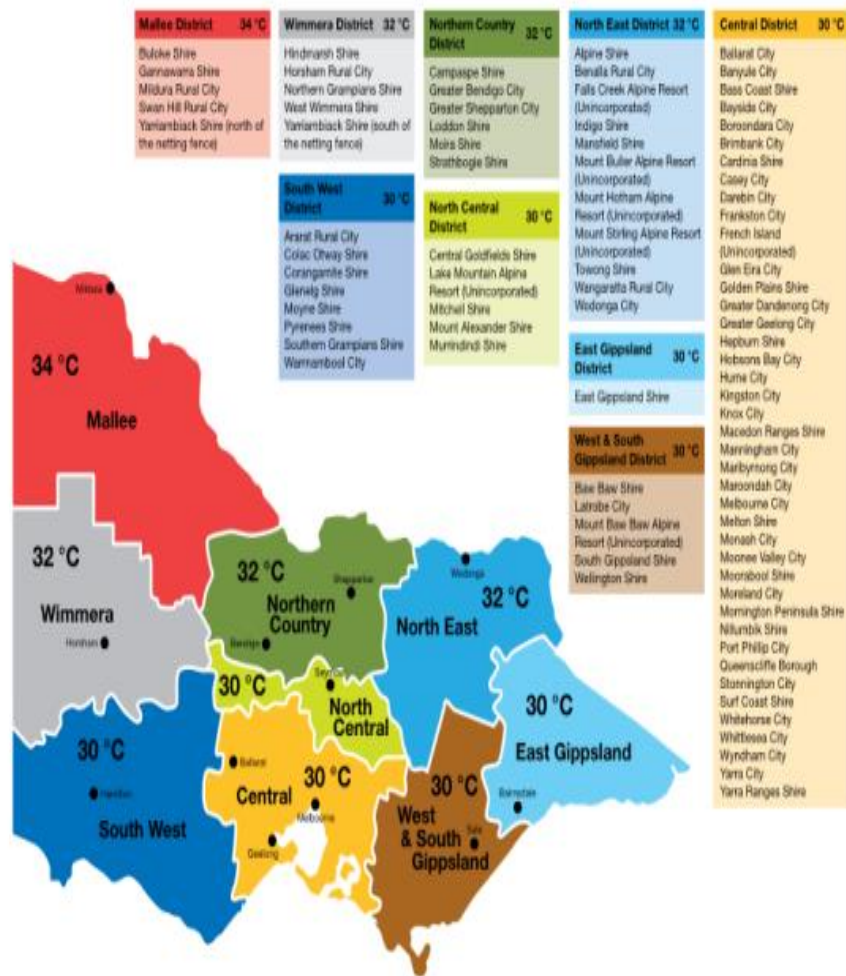
Max: 31°C

AVERAGE CALCULATION FOR TUESDAY $(45 + 26) / 2 = 35.5^{\circ}\text{C}$

The threshold for the Indigo Shire = mean of 32°C (see figure 1 below), therefore, the temperature forecast indicates that the threshold will be exceeded.

The above calculation should be used to calculate the mean temperature of everyday within a seven-day forecast which is available on the BOM website:

[Indigo Valley Forecast - Bureau of Meteorology](#)



Source: <https://files-em.em.vic.gov.au/public/EMV-web/SERP-StateExtremeHeatSub-plan.pdf>

Figure 1: Heat Health temperature thresholds by weather district

7 HOW HEAT WAVES AFFECT OUR HEALTH

As temperatures rise, so does the risk of a heat related illness, a medical condition that results from the body's inability to cope with heat and cool itself. If left untreated, a heat illness can lead to serious complications, even death. Heat kills more people than fires, tornadoes, hurricanes, lightning and flash floods – combined.

Heat related illness can make people feel uncomfortable, not so much because they feel hot, but rather because they sense how difficult it has become to lose body heat at the rate necessary to keep their inner body temperature close to 37°C. The body responds to this stress progressively through three stages.

STAGE	SYMPTOMS	WHAT TO DO
1 - Heat cramps	Muscular pains and spasms in the abdomen, arms or legs.	Stop activity and sit quietly in a cool place, increase fluid intake, rest a few hours before returning to activity, seek medical help if cramps persist.
	This is an early signal that the body is having trouble with the heat when the body gets depleted of salt and water	
2 – Heat exhaustion	Cold, clammy and pale skin and sweating, rapid heart rate, muscle cramps, weakness, dizziness, headache, nausea, vomiting, fainting.	Get the person to a cool area and lay them down, remove their outer clothing, wet their skin with cool water or wet cloths, seek medical advice.
	Blood flow to the skin increases, causing a decrease of flow to the vital organs. This results in mild shock. If not treated the victim may suffer heat stroke.	
3 – Heat stroke	Same as heat exhaustion with – dry skin with no sweating, mental condition worsens, confusion, seizure, stroke-like symptoms or collapsing, unconsciousness.	Call an ambulance immediately, get the person into a cool area and lay them down, remove their clothing, wet their skin with water, fanning continuously. Position an unconscious person on their side and clear their airway.
	This is life threatening. The victim's temperature control system, which produces sweating to cool the body, stops working. The body temperature may exceed 40.6°C potentially causing liver, kidney, muscle, heart, brain damage and death if the body is not cooled quickly.	

High temperatures are linked to:

- An increase in hospital admissions relating to heat stress, dehydration, or as a result of heat exacerbating existing conditions;
- Increased rates of certain crimes particularly those related to aggressive behaviour such as homicide;
- Increased number of work-related accidents and reduced work productivity; and

Prevention

To help prevent the onset of a heat related illnesses, people in the Indigo Shire are advised to:

- Monitor the weather forecast and the Bureau of Meteorology Heatwave warnings online or via the Bureau's app. Subscribe to receive the Department of Health Heat Health warnings.
- Use air-conditioning if available. The cost of air-conditioning can be reduced by using a fan at the same time
- Electronic Fans can help cool the body when the indoor temperature is below 39-40 degrees C
- Keep skin wet using a spray bottle or damp sponge or cloth
- Soak a towel in cool tap water and wrap it loosely around your head
- Wrap ice cubes in a damp towel and drape around your neck
- Wear light and loose-fitting clothing
- Consider visiting an air-conditioned building such as a shopping centre or public library
- Use blinds or curtains to block the sun from shining directly through windows
- Open windows and doors if you think it is hotter indoors than outdoors
- Stay hydrated, in extreme heat keep drinking water before you feel thirsty, especially if outdoors or performing physical activity.
- Check in with others. Older people or otherwise vulnerable people with medical conditions are more susceptible to heat-related problem

General care for heat emergencies includes cooling the body, giving fluids and minimising shock.

8 RECOMMENDED COOL RELIEF SPACES

In the event of a heatwave, it is recommended the public source cool areas. Easily accessible areas include public swimming pools, libraries, some community centres, local shopping centres and other areas. The council may identify and consider opening community spaces for people to attend to escape from the heat and may consider extending pool operating hours. Recommended cool areas, including the local pool opening hours, will be advertised by the council through the council website, the local newspaper, radio and other social networks. Swimming Pools will not be available on declared catastrophic days.

On high heat days, it is recommended taking a day trip to the nearest regional city, where there are places to stay out of the heat such as shopping centres, cinemas etc. and you can remain in touch with family and friends.

On days declared as a 'Catastrophic' Fire danger day council facilities will be closed therefore, you should have a plan as to how you are going to manage staying cool however, if you reside in a high-risk area you should consider leaving and going to somewhere that is not only safer but has the capacity to keep you cool until the risk passes.

Indigo Shire and Beechworth Health Service recognises a number of places and building available for the community to take refuge in during days of extreme heat.

A full list of 'Cool Relief Spaces' can be found at;

<https://www.indigoshire.vic.gov.au/Residents/Emergencies/Heatwaves/Cool-Relief-Spaces>



9 PEOPLE MOST AT-RISK TO HEATWAVE CONDITIONS

Although anyone can suffer from the effects of prolonged periods of heat at any time, some people are at greater risk than others. Those most vulnerable include the elderly, infants and young children, people with a disability, people with a pre-existing medical condition and people who are socially and economically disadvantaged. Other groups at risk to heat include people who are overweight, people who overexert during work or exercise and people confined to bed and unable to care for themselves. While the *Indigo Shire Extreme Heat Plan* plans for a whole-of-community response, it particularly focuses on those aged above 65 years of age, children between 0 and 4 years of age, people with chronic disease, those with a disability, on medication or working outdoors, tourists and farmers.

Our elderly

Elderly people in our community are more prone to heat stress for several reasons.

- Elderly people do not adjust as well as young people to sudden changes in temperature.
- They are more likely to have a chronic medical condition that upsets normal body responses to heat.
- They are more likely to take prescription medicines that impair the body's ability to regulate its temperature or that inhibit perspiration.

Indigo's population, like most parts of Australia, is ageing in 2021, the median age of people living in Indigo was 47 years. Of the 17,368 residents in Indigo in 2021:

- 8.3% were aged 60-64 years
- 7.8% were aged 55-59 years
- 7.5% were aged 65-69 years.

With a median age of 32 years, Tangambalanga is the youngest locality in Indigo. The oldest is Stanley (Vic.) with a median age of 54 years.

70-74 years cohort - recorded the largest change in Indigo, with an additional 363 people, a 44.6% increase from 2016.

Our young children

Young children are sensitive to the effects of high temperatures because they produce more metabolic heat than adults and rely on others to regulate their environments and provide adequate liquids. Their core temperature can rise quickly causing dehydration. We need to watch our children carefully in times of prolonged heat as they can succumb to heat illness in a very short period.

Although numbers are projected to fall in the next few years, we still have a significant share of families with young children in our shire. At present children aged 0 to 4 years comprise 4.89% of our total population. They are catered for by six Maternal and Child Health centres located across the shire and have access to a range of children's services including childcare centres, family day care, kindergartens, preschools and playgroups.

People with chronic disease

Virtually all chronic diseases present a risk of death/illness due to heat. The Indigo shire has a high rate of some chronic diseases when compared with the rest of Victoria.

The leading main disease groups contributing to the burden of disease in Indigo Shire include:

- Cancer,
- Cardiovascular disease,
- Dementia; and,
- Diabetes.

Other groups with chronic disease at increased risk of illness and death associated with heat include people with mental illness. Dehydration can cause stress on the heart, known as cardiac stress.

The prevalence of chronic disease and modifiable chronic disease risk factors have increased over time, in conjunction with increases in life expectancy. As the population ages, the number of people with a chronic disease in the Indigo shire is expected to increase. This presents a significant challenge to the health system with important implications for the future health and wellbeing of the population in periods of extreme heat.

People with a disability

Some people with a disability can be affected by heat because their body may not be able to regulate body temperature. This means the body may not be able to lose heat through skin by sweating or by having blood flow to the surface of the skin which helps the body to cool down. Also, excess fat or wearing too many clothes can cause heat stress. Dehydration or not drinking enough water also can cause heat stress as there may not be enough fluid in the body to facilitate sweating.

A person with cognitive impairment, whether from disease or injury, may not be able to communicate distress or need for water. In some cases, they may not even “feel” the heat or discomfort because of changes in the brain’s abilities to process sensory information or regulate their body’s responses to heat.

People on medication

Some medications increase the risk of heat stress. This varies according to the medication, for example:

- Antidepressants, antihistamines (allergy medicines), anticonvulsants (seizure medicines), phenothiazine and anticholinergics (used for some psychiatric conditions) act on an area of the brain that controls the skin’s ability to make sweat.
- Beta blockers and vasoconstrictors (heart tablets) reduce the ability of the heart and lungs to adapt to stresses including hot weather.
- Amphetamines (stimulants) raise the body’s temperature.
- Diuretics (fluid tablets) act on the kidneys and encourage fluid loss. This can quickly lead to dehydration in hot weather; and,
- Opioids and sedatives can reduce the person’s awareness of physical discomfort, which means symptoms of heat stress may be ignored.

People who are outdoors

Many people in Indigo Shire work or play outdoors in hot conditions, irrespective of the weather. Air temperature, radiant temperature, air velocity, humidity, clothing and activity are recognised as factors that interact to determine heat stress. Anyone having to work or play outside in hot weather without appropriate protection, particularly if this involves heavy physical activity, is at increased risk of suffering health effects from heat. Therefore, certain occupational and sporting groups need to be informed about possible measures to prevent heat stress, how to recognise heat stress, heat exhaustion and heatstroke, and what to do. Effective management systems for ensuring the health and safety of workers and sports people (especially children) must be in place.

Messaging for people who are outdoors that on very hot days they should follow the advice on avoiding heat and;

- When it’s hot, it’s time to slow down and drink lots of water;
- Wear a broad-brimmed hat and cool loose clothes, long sleeve shirts which “breathe”.
- Use sunscreen and carry copious supplies of water (drink around 500 ml before leaving your accommodation, then 200-300 ml every 15 minutes); and
- Reduce exercise in duration and intensity, or postpone to a more suitable, cooler time.

Our tourists

In heat events, dehydration and heat stress can be a risk, particularly for non-acclimatised people. North East Victoria attracts 1.3 million domestic overnight visitors per annum, many of them in the summer months. Whilst the Indigo Shire is recognised as a safe place to visit, as visitor numbers increase over summer, additional pressure can be placed on the capacity of health and safety services to meet the needs of visitors if they suffer from heat related illnesses.

Our farmers

Prolonged periods of hot weather can have far-reaching effects on our farmers. Because farmers generally work outdoors, they are at an increased risk of heat stress. Research tells us that during heatwave

conditions there are higher numbers of work-related farm accidents and reduced work productivity, yet increased workloads and higher levels of stress for farming families. Deterioration of water supplies, along with drought and bushfire threat, can further add to difficult and expensive summers for our farmers.

Agriculture contributes to the shire's economic base and employs around 722 persons. There are approximately 2,100 rural rated properties in the Indigo Shire, however, many are less than 200 hectares in area and are small, hobby farms. Some of these farming communities are situated in isolated geographic locations away from services and supports.

Likely impacts

The risks for all vulnerable groups during heatwaves are dehydration and heat related illness. Of particular concern during days of high heat is for those living alone and/or disconnected from services and networks, and increased workloads and stress levels for health service staff.



10 OTHER EFFECTS OF EXTREME HEAT

Extreme heat rarely occurs in isolation. Infrastructure failure or other natural emergencies can add another level of demand on a community and services. For example, power outages will impact on people's ability to run air-conditioners; bushfires will increase vulnerability by reducing air quality; and public transport disruptions will hinder people's ability to reach a cooler location.

Power

Heat events increase the demand for electricity and can cause infrastructure failure or require demand be controlled by power shedding. The loss of power can impact on the use of all cooling mechanisms including fans and air conditioning, the ability to keep food stored at hygienic levels and the ability to provide water and household use where a property is reliant on electric pumps.

Electricity distributors such as AusNet Services register addresses of power dependent people (life support customers). Customers who depend on electricity for life support equipment must inform their supplier to ensure immediate relief is provided.

Water

Heat events can cause an increase in the demand for water services, particularly drinking water. Concurrent risks can occur to distribution and treatment when power supply is also lost. Loss of the reticulated supply system can impact on use of evaporative air conditioning to stay cool, sanitation services and access to drinking water.

Animals

Animals must be cared for in a heat event. The *Indigo Shire Municipal Emergency Animal Welfare Plan* outlines the responsibilities, interactions and activities provided to ensure animals are cared for during days and extended periods of high heat.



11 OUR PARTNERS

The Indigo Shire Council in partnership with the Emergency Management Commissioner (the Incident Control agency in a heatwave event) will provide key messages and important information on heatwaves, consistent with the Chief Health Officer's public health messages and communication resources to residents over the summer period. The council outlines how these messages are distributed, disseminated and worded in the *Extreme Heat Action Plan* – Appendix 2

Council is working alongside stakeholders from within the targeted sector groups in the Indigo community to act as key partners in the *Extreme Heat Plan*. We have partnered with government, private sector, service provider and community group organisations that represent especially our affected vulnerable communities of the very young, elderly, those with chronic disease, a disability, on medication or working outdoors, tourists and farmers. These partners include:

• Beechworth Health Service • Emergency Services (Vic Pol, SES, CFA, Ambulance Vic & Redcross)	• Department of Veteran affairs • Department of Energy, Environment and Climate Action (DEECA)
• Indigo North Health Service	• Educational institutions
• Yackandandah Health	• Beechworth Visitor Information Centre
• Gateway Health	• Chiltern Visitor Information Centre
• Indigo Health Consortium • Sporting groups	• Yackandandah Visitor Information Centre
• Local preschools, playgroups and Childcare Centres • Department of Families Fairness and Housing (DFFH) • Goulburn Broken Greenhouse Alliance • North East Water	• Rutherglen Visitor Information Centre • Upper Hume Primary Care Partnership • Ausnet services • Community service providers (Senior Citizens and Salvation Army)
• Ovens Murray Local Public Health Unit	• Telstra/Optus/NBN
• Goulburn Murray Water	• Parks Vic

12 WHAT WE ARE GOING TO DO

To be better prepared for and responding to extreme heat conditions during the summer months the Indigo Shire Council is going to:

- Include extreme heat preparation, response and recovery into existing municipal plans;
- Make use of existing community registers;
- Promote the use of cool areas in key locations around the shire;
- Use our community services and organisations to support vulnerable populations;
- Engage in a communication and media campaign using heatwave messages consistent with Department of Families Fairness and Housing and Department of Health materials; and
- Respond to state activated heat alert system in a planned and considered way.

Our action plan

Our action plan is not just about responding to a pending extreme heat; instead, it provides guidance all-year-round as we prepare our community in advance for very hot summers. The action and communications plans have clearly identifiable stages that include a set of clear actions (refer to appendix 2).

Our actions then can be divided into five stages.

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5
(Before)	(Before)	(Before)	(During)	(During)
Pre- Summer preparation (April 1 to November 30)	During Summer Prevention (December 1 to March 31)	Alert (0-7 days before heat event)	Heatwave response	Recovery and review

Each stage is characterised by a set of key actions

Stage One Pre summer preparation April 1 to November 30	• Implement Extreme Heat Awareness Campaign (April to Nov) • Identify vulnerable groups and update community registers • Identify cool areas and access to water • Engage key stakeholders
Stage Two During summer prevention December 1 to March 31	• Implement Extreme Heat Awareness Campaign (Dec to March) • Advise key stakeholders of roles and responsibilities • Organise cool areas for possible use • Monitor Bureau Of Meteorology thresholds for Indigo Shire
Stage Three (0-7 days prior to heat event) Preparation phase – Based on the 7-day weather forecast	• Promote (and advise staff of their obligations under) hot work policies and procedures throughout Council • Promote and provide key heat health messages to stakeholders • Engage key stakeholders • Monitor local conditions
Stage Four Heatwave response Trigger: DH heat alert	• Alert key stakeholders to enact specific actions • Ensure HAC provider (Alpine Health) contacts residents that are on VPR and makes them aware of heatwave risks • Advertise cool areas • Inform Emergency Management Team • Monitor Bureau of Meteorology reports
Stage Five Recovery and Review	• Debrief with staff to improve this plan and local procedures • Implement improvements to this plan following feedback • Review hot work policies and procedures and make improvements as required

Further details of how these actions are enacted are found in Appendix 2 – *Extreme Heat Action and communications Plan*.

Implement the Heatwave Awareness Campaign

The campaign will provide key heat/health messages to the community during this time via council's regular communication channels. These messages will include:

COOL YOUR HOME DOWN	• Keep windows that are exposed to the sun closed during the day, and opened at night when the temperature has cooled • Turn off non-essential lights and electrical equipment • Move to the coolest room to sleep
STAY OUT OF THE HEAT	• Keep out of the sun during the hottest part of the day • Avoid extreme physical exercise • Wear light, loose-fitting clothes • Reschedule appointments to early morning • If outdoors wear appropriate clothing including; hat, long sleeve shirts or appropriate clothing and apply sunscreen
KEEP YOURSELF COOL AND HYDRATED	• Drink plenty of water; avoid caffeine/ alcohol • Take a cool shower • Spray water over your skin or clothing • Keep a damp cloth on the back of your neck
WHAT YOU CAN DO FOR OTHERS	• Phone or visit elderly or sick neighbours • Be aware of phone numbers you can ring to get help • Know where cool areas are in the shire

Advise key stakeholders of roles and responsibilities

Council will contact key stakeholders to advise them on the implementation of Stage 2 of their plans. Sporting bodies and summer event organisers will also be informed of safe heat thresholds.

Organise cool areas for possible use

During Stage 2, Council will ensure that designated cool areas within the shire are ready for use in the event of a heatwave. Shire public pools will be informed of Stage 3 actions.

Monitor Bureau of Meteorology thresholds for Indigo Shire

Indigo Shire Council will check Bureau of Meteorology reports daily for weather updates.

13 HOW WE EVALUATE THE PLAN'S EFFECTIVENESS

To ensure Indigo's *Extreme Heat Plan* remains relevant and meets the changing needs of our community, Council will review it annually after each summer and use any amendments to inform an updated *Extreme Heat Plan* every four years. All relevant stakeholders will actively participate in the evaluation. The following questions will be addressed.

- Were the actions in Stage 3&4 appropriate and timely?
- What worked?
- What didn't work?
- Was information communicated effectively to stakeholders?
- Was information communicated effectively to the general community?
- What could we do differently next summer?

The overall objective of the Extreme Heat Plan is to ensure that death and serious injury is prevented during an extreme heat event.

14 Appendices

Appendix 1: Key terms

Bureau of Meteorology: The Bureau of Meteorology provides trusted, reliable and responsive weather, water, climate and ocean services for Australia. The Bureau of Meteorology web homepage provides the Australian community with access to weather forecasts, severe weather warnings, observations, flood information, marine and high seas forecasts and climate information.

Climate: Climate summarises the average, range and variability of weather elements (rain, wind, temperature, fog, thunder and sunshine) observed over many years at a location or across an area. It includes the future expectation of long-term weather, in the order of weeks, months or years ahead.

Climate change: Climate change is the change in average weather over time and over a region. Climate change includes changes in temperature, wind patterns and precipitation.

Department of Health Victoria: The Department of Health was established to help Victorians stay safe and healthy and deliver a world-class health system that leads to better health outcomes for all Victorians. Responsibilities include supporting the Health and Ambulance Services, Mental Health and Ageing portfolios.

El Niño: El Niño refers to the extensive warming of the central and eastern Pacific that leads to a major shift in weather patterns across the Pacific. In Australia (particularly eastern Australia), El Niño events are associated with an increased probability of drier conditions.

Global warming: Human activities have caused the warming of the global climate over the last 150 years, accompanied by retreating alpine glaciers, rising sea levels and shifting climate zones. Scientists believe that global warming will lead to changes in wind patterns, precipitation and frequency and type of severe weather events. This will have significant environmental and economic consequences.

Greenhouse effect: The greenhouse effect is a natural warming process of the earth. Without greenhouse gases, the earth's average surface temperature would be about 35 ° Celsius cooler. Scientists are concerned that higher greenhouse gas concentrations, resulting from human activity, will lead to an "enhanced" greenhouse effect which will lead to global climate change.

Extreme Heat: Is a period of abnormally hot weather lasting several days.

Morbidity: Morbidity is an incidence of ill health. It is measured in various ways, often by the probability that a randomly selected individual in a population at some date and location would become seriously ill over a period.

Mortality: Death.

Vulnerable Persons Register: The VPR provides personal care, support, and case management services across Victoria to people living in the community. The collaboration and coordination with local governments creates the ability for the police to access key information about individuals allowing for quick and effective emergency responses.

Appendix 2: Heatwave action & communications plan

STAGE	ACTION	HOW	WHEN	RESPONSIBLE PERSON
Stage One – Pre-summer preparation April 1 to November 30	Review Extreme Heat Plan	Review plan and action plan	September	MRM, MEMO & EMC
		Review heat health alert protocols for workplace supervisors and staff, and educate staff		All managers
	Does this need to be DH Heat Campaign?		October	MRM, EMC & Communications
		Review DH website		
		Order or restock resources		
	Implement Extreme Heat awareness	Distribute resources to key locations and restock heat health information: • Childcare centres • Maternal & child health centres • Preschools • Visitor information centres • Libraries • Sporting facilities • Swimming pools • Senior citizens	October	EHO
	Coordinate pre-prepared media releases (April-Nov)	Review and prepare media releases and pre-prepared key heat messages	April - Nov	Communications
		Develop/review FAQ on council website		
	Identify vulnerable groups and update community register	Update Vulnerable persons register	April - Nov	MRM
		Ensure all persons on the VPR are aware of the activation of stage one		
	Identify Cool areas & water access	Council to identify and consider community spaces to promote to the general public as cool areas during the awareness campaign	April - Nov	MRM
		Spread keep cool messages • Indigo Informer • Seniors celebrations • Children's week • Community noticeboards		Communications

		Indigo Shire website		
	Engage key partners	Council to advise stakeholders (identified in our partners) to revise and amend key heat health messages and actions such as: - Review their heat alert plans - Update their community registers - Pre-prepare key heat messages - Restock heat health information in relevant places - Assist vulnerable persons to include heat health planning in personal emergency management plans	April - Nov	MRM & EMC
Stage two – During Summer prevention December 1 to March 31	Implement Extreme Heat Awareness Campaign (Dec-March)	Activate Council's Extreme Heat Awareness Campaign: - Media releases - Identify spokesperson	Dec - Mar	Communications
	Advise key partners of roles and responsibilities	Advise key stakeholders to implement stage two of their plans	Dec	MEMP Committee and EMC
	Organise cool areas for possible use if applicable	Activate and organise cool areas if applicable and publicise	Dec	MRM, EMC & Communications
	Vulnerable People	Ensure all persons on VPR are aware of the activation of stage two by contacting responsible case management agencies	Nov	MRM
	Monitor BOM reports for Indigo Shire	Monitor BOM for daily weather updates and thresholds for Indigo	Dec - Mar	MRM
	Heat Health Warning	Council advised of Heat Health warning		Chief Medical Officer - DH
		Reschedule relevant staff work hours		All managers
Stage three - Alert (0-7 days before heat event)	Promote hot work policies and procedures throughout Council	Advise staff of their obligations under hot work policies and procedures		All managers
	Promote, coordinate and oversee implementation of extreme heat action	Council to advise all internal stakeholders to implement of the heatwave plan. Including:		MEMO

	plan following heat health warning trigger	• Prioritising tasks and clients • Assessing risk for clients and staff • Rescheduling staff work times and hours • Providing additional fluids for clients and staff • Modifying opening hours of any council cool relief places • Cancel any council outdoor events Council will consider undertaking some or all the following (external stakeholders). Including: • Council website updated with current information regarding safety messages and cancellation/ alterations to Council services • Media releases • Social media messaging • Develop media and communications messages that inform the community about: - Possible changes to normal Council service delivery - Basic safety measures - Information on where to get further information and/ or medical advice • Ensure all persons on VPR are contacted by HAC provider and made aware of alert • Ensure HAC provider is in regular contact with vulnerable clients		Communications MRM & Alpine Health
	Vulnerable clients			

	Advise key partners to enact specific actions	Council to advise all external stakeholders to implement their heatwave plan. This may include: - Altering children's outdoor activities and playtimes - Altering or cancelling sports training and events - Modifying client programs		MEMO
	Promote Cool areas and key heat health messages	Social Media		Communications Library
	Inform emergency management team			MEMO
	Monitor BOM and Heat Health warnings	Continue to monitor BOM for daily weather updates and thresholds for Indigo and messages from Chief Health Officer at Department of Health.		MRM and Communications
Stage four – Heat Health response. Trigger: Chief Health Officer heat alert	Vulnerable clients	- Ensure all persons on VPR are contacted by HAC provider and made aware of alert - Ensure HAC provider is in regular contact with vulnerable clients		MRM & Alpine Health
	Advise key partners to enact specific actions	Council to advise all external stakeholders to implement their heatwave plan. This may include: - Altering children's outdoor activities and playtimes - Altering or cancelling sports training and events - Modifying client programs		MEMO
	Promote Cool areas and key heat health messages	Social media		Communications Library
	Inform emergency management team			MEMO
	Monitor BOM and Heat Health Warnings alerts	Continue to monitor BOM for daily weather updates and thresholds for		MRM and Communications

		Indigo and messages from Chief Health Officer at Department of Health.		
	Monitor BOM and Health alerts	Continue to monitor BOM for daily weather updates and thresholds for Indigo and messages from Chief Health Officer at Department of Health.		MRM and Communications
Stage 5 - Post event Recovery and review	Evaluate	• Review • Revise • Replenish, restock heat health information in relevant places		MEMO, MRM, EMC & Communications
	Debrief with staff to improve this plan and local procedures Implement improvements to this plan following feedback Review hot work policies and procedures and make improvements as required		After event	MEMO, MRM, EMC, managers

Appendix 3 – Flows Chart for Communications Plan

